

Dhyey Patel

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Data Scientist with an MS in Data Science, skilled in data analysis, visualization, predictive modeling, time series analysis, NLP using Python, SQL, BI tools, ML models, AWS and more. As a new graduate, I bring strong analytical abilities, adaptability to new tools, and a passion for continuous learning to real-world challenges, collaborate with cross-functional teams, and contribute to meaningful business outcomes. Motivated to grow into a leadership role, I am committed to developing technical expertise and delivering long-term value to the organization.

TECHNICAL SKILLS

Programming Languages: Python, SQL, R

ETL tools and Databases: MySQL, Neo4j Graph, MongoDB, Snowflake, Databricks

Data Visualization Tools: Tableau, Power BI, Looker, MS Excel

Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, Keras, TensorFlow, NLTK, PyTorch, BeautifulSoup, PySpark

ML Algorithms: Linear/Logistic Regression, Naïve Bayes, Random Forest, XGBoost, LightGBM, SARIMA, Neural Networks

Methodologies: Data Analysis, Visualization, Dashboards, Reports, Predictive modeling, Forecasting, ETL, NLP, A/B Testing

PROJECTS

E-commerce Review Sentiment Analysis for Strategic Marketing Decisions

- Scraped millions of product reviews from e-commerce platform like Amazon using **pagination** and **batch processing** techniques, ensuring efficient data collection from multiple pages without overwhelming the server. Extracted reviews, ratings, and metadata for sentiment analysis using **Natural Language Processing (NLP)**.
- Applied sentiment analysis using **TextBlob** and **VADER** to classify reviews as positive, neutral, or negative. Analyzed the sentiment distribution to gauge public opinion on specific brands and products, helping **marketing teams** refine their strategies based on customer feedback. **Visualized sentiment trends** over time and across **product categories**.

Fraud Detection Using Graph Neural Networks

- Developed a **fraud detection solution** leveraging **Graph Neural Networks (GNN)** like **GraphSAGE** to classify fraudulent activities in transactional data. Modeled relationships among **millions of transactions** as a graph, achieving a **significant improvement** in **detection rates** and reducing false negatives by 15%.
- Performed **exploratory analysis** and **clustering** using **DBSCAN** to identify unusual **transaction patterns**. Integrated the system into **Neo4j** for **real-time fraud detection**, improving response times and securing financial operations.

Twitter Bot Detection and Behavior Analysis

- Leveraged the Twitter-Bot Detection Dataset to build a **prediction model** for identifying bot users. Conducted **data preprocessing**, **feature engineering**, and **hyperparameter tuning** using algorithms like **Random Forest** and **Neural Networks**. Combined the outputs from both models through **weighted approach** to mitigate individual **model biases**.
- Designed and implemented a **hybrid forecasting model** by integrating **XGBoost** for high-accuracy **short-term predictions** and **Prophet** for robust **long-term trend forecasting**. Achieved **85% accuracy** in predictions, optimizing performance with a blend of machine learning and statistical methods.

WORK EXPERIENCE

Jain Alert Group USA

Jun 2023-Aug 2023

Data Analyst Intern

- Performed in-depth analysis of data from multiple events and programs using Python and SQL to identify actionable insights, enhancing event targeting strategies and boosting volunteer engagement by 15%.
- Developed dashboards in Tableau and MS Excel to visualize trends and optimize fundraising efforts, contributing to a 40% improvement in donor retention and an increase in overall donations.
- Supported marketing and operations teams by implementing data-driven decision-making processes to improve event targeting and management strategies.
- Tracked KPIs such as donation growth rate and event attendance rate for campaign performance, enabling real-time adjustments for maximum impact.

Silverwing Technologies Pvt. Ltd.

Oct 2021-May 2022

Python Intern

- Developed a web application using Python (Django), HTML, and CSS to provide detailed information on towed vehicles, including reasons for towing, fines, towed location, and real-time vehicle tracking with Google Maps API.
- Addressed common driver issues like flat tires and engine overheating using statistical analysis to identify root causes, reducing roadside assistance calls by 40% and improving customer satisfaction ratings.
- Designed dashboards in Tableau and generated reports with SQL and Excel to analyze patterns such as high-risk breakdown areas and peak towing times, enabling actionable insights to improve operational efficiency and reduce response times.

EDUCATION

Pace University, Seidenberg School of Computer Science and Information Systems

Sep 2022-May 2024

Master of Science (MS) in Data Science

Gujarat Technological University

Aug 2018-May 2022

Bachelor of Engineering (BE) in Information Technology